VayuSync — Hyperlocal Stubble Exposure & Institutional Action Engine

Track: Track 1 (Air) | **Target:** ₹2,00,000 Track Winner & Amazon Fast-Track Interview Shortlist
Event: Bharat Builds Tour 2026: Environmental Hacks

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1. Executive Summary & Problem Framing

1.1 The Visualization Trap (Why 90% of Air Track Projects Fail)

Most environmental hackathon entries build reactive AQI dashboards. They pull static sensor data from government feeds, render red-orange-yellow heatmaps, and display historical trend graphs. By the time a central AQI monitor records a hazardous spike of 450+ PM2.5 in Delhi NCR, the toxic smoke plume has already settled over high-density population centers for hours.

Passive visualization provides **zero operational lead time**. It forces institutions to react after exposure has occurred, rendering interventions like masking or HVAC adjustments ineffective against peak toxic inhalation.

1.2 The Core Problem

During October and November, agricultural stubble burning across Punjab and Haryana releases massive thermal anomalies and particulate plumes. Driven by seasonal north-westerly wind vectors, these plumes travel across regional corridors and hit vulnerable urban micro-zones in Delhi NCR.

Vulnerable institutions lack predictive micro-targeting:

- **Schools:** Learn about severe air quality spikes during mid-day sports or outdoor assemblies, hours after exposure.

- **Outdoor Workforces:** Construction sites and municipal sanitation crews operate without automated schedule adjustments or shift rotations prior to plume arrival.
- **Healthcare Facilities:** Hospital administrators cannot dynamically scale indoor air filtration or prep respiratory wards ahead of surge arrivals.

1.3 The VayuSync Solution

VayuSync shifts the paradigm from *reactive tracking* to *preemptive institutional automation*. Built as an autonomous, event-driven cloud pipeline on AWS, VayuSync tracks real-time thermal anomalies from satellite data, calculates downwind smoke trajectories using live wind vectors, and automatically triggers targeted operational guardrails **6 to 12 hours before** toxic smoke reaches specific micro-zones.[NASA FIRMS Thermal Data] + [Open-Meteo Wind Vectors]



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2. Core Project Goals & Success Metrics

2.1 Architectural & Operational Goals

- **Goal 1: Zero-Maintenance Serverless Pipeline**
  Deploy a fully serverless, event-driven infrastructure on AWS SAM/CDK with zero persistent server management, minimal warm-up latency, and zero cold start penalty on critical alert paths.
- **Goal 2: Hyperlocal Downwind Risk Prediction**
  Fuse NASA FIRMS satellite fire hotspot data, Open-Meteo 10m wind direction/velocity vectors, and ground-level CPCB sensor streams to compute high-resolution dispersion trajectory cones.
- **Goal 3: Actionable Institutional Guardrails**
  Utilize Amazon Bedrock (Anthropic Claude 3 Haiku) to evaluate calculated exposure vectors against registered institutional thresholds, generating deterministic, safety-checked operational directives.

2.2 Key Performance Indicators (KPIs)

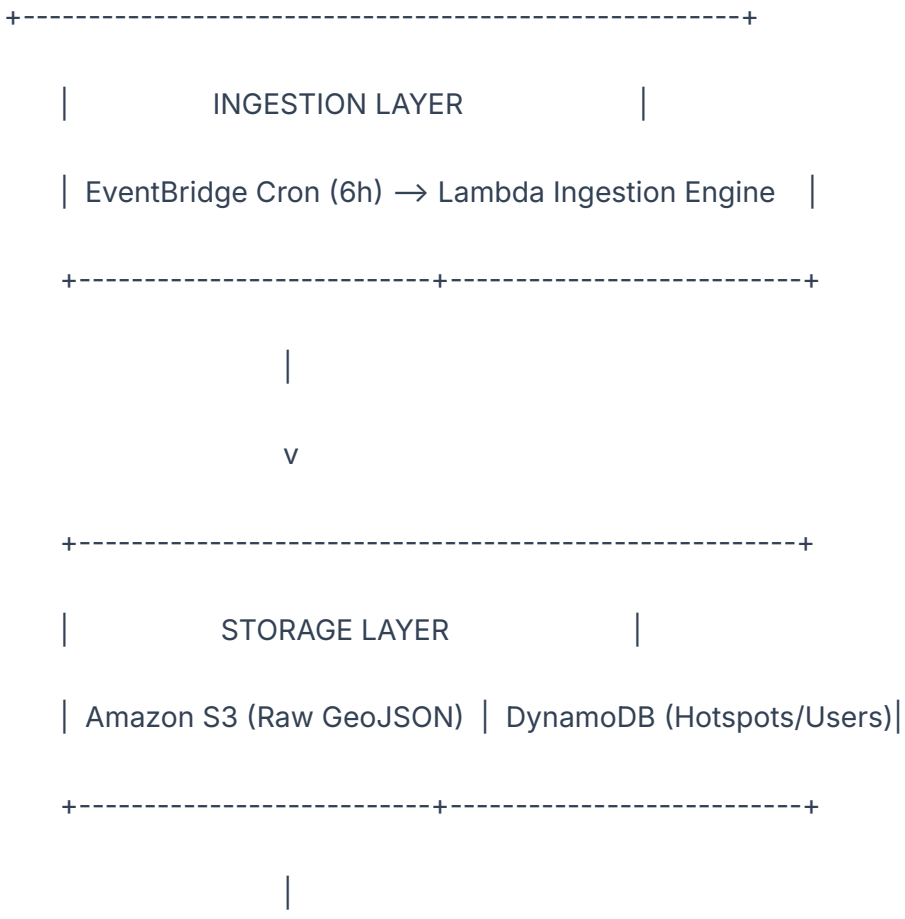
Performance Metric	Operational Target	Validation Method
End-to-End Processing Latency	< 2.5 seconds	Timestamp logs from EventBridge trigger to SNS dispatch
Alert Dispatch Determinism	100% execution success	CloudWatch Logs verification for mock WhatsApp/SMS hooks
AWS Cost & Quotas	100% Free Tier Compliant	AWS Budgets alarm configured at $0.00 threshold

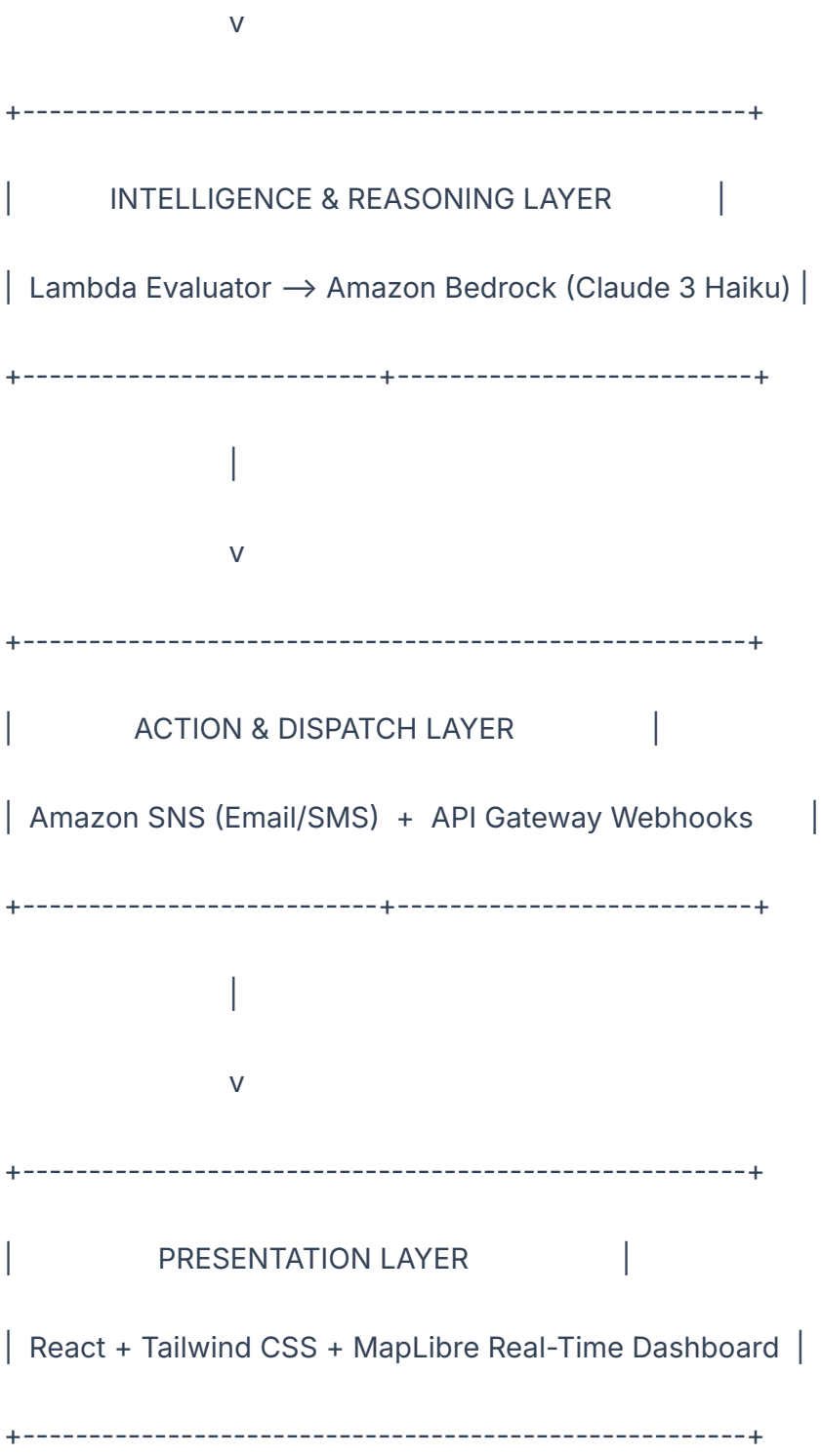
Performance Metric	Operational Target	Validation Method
Predictive Lead Time	6–12 Hours	Downwind vector calculation ahead of surface sensor spikes
System Availability	99.9% Serverless Uptime	AWS Lambda & DynamoDB multi-AZ native durability

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3. AWS Cloud Architecture & Data Pipeline

VayuSync utilizes an event-driven, decoupled AWS serverless architecture designed for scale, rapid execution, and complete cost efficiency.





3.1 Data Ingestion Layer

- **Amazon EventBridge:** Executes a scheduled rule every 6 hours (and on-demand demo triggers) to invoke the ingestion orchestrator.

- **AWS Lambda (`VayuSync-IngestionEngine`):**
    - Fetches active thermal anomalies from NASA FIRMS (VIIRS/MODIS C2 data feed for South Asia).
    - Queries Open-Meteo API for real-time and forecasted 10m wind vector components (u-wind, v-wind) and boundary layer height across Punjab, Haryana, and Delhi NCR coordinates.
    - Standardizes raw multi-source payloads into normalized GeoJSON structures.

3.2 Storage Layer

- **Amazon S3 (`vayusync-raw-payloads-2026`):** Stores raw GeoJSON responses from external APIs for historical auditability, model verification, and replay testing.
- **Amazon DynamoDB (`VayuSync_Hotspots`):** Stores processed hotspot records with spatial attributes, confidence metrics, estimated PM2.5 mass fluxes, and TTL (Time-To-Live) set to 48 hours for cost management.
- **Amazon DynamoDB (`VayuSync_Institutions`):** Contains registered vulnerable targets (schools, hospitals, outdoor sites) with geofenced coordinates, operational hours, contact channels, and risk sensitivity thresholds.

3.3 Intelligence & Reasoning Layer

- **AWS Lambda (`VayuSync-Evaluator Engine`):**
    - Executes a kinematic lagrangian dispersion algorithm to project 6-hour and 12-hour downwind risk trajectories (dispersion cones) using wind vectors.
    - Queries `VayuSync_Institutions` using spatial bounding boxes to identify at-risk locations within projected impact zones.
- **Amazon Bedrock (Anthropic Claude 3 Haiku):**
    - Receives structured JSON input containing calculated arrival times, local meteorological metrics, and specific institutional profile data.
    - Generates structured, highly specific, and actionable mitigation advisories while enforcing strict system prompts to prevent hallucination or unstructured output.

3.4 Action & Dispatch Layer

- **Amazon SNS (`VayuSync-AlertTopic`):** Broadcasts critical push notifications via SMS and Email to institutional administrators.
- **API Gateway + Lambda (`VayuSync-WebhookDispatcher`):** Fires JSON webhooks to simulated automated building management systems (BMS) to automatically switch HVAC units to recirculate mode or trigger automated WhatsApp notification flows.

3.5 Presentation Layer

- **Single Page Application (React + Tailwind CSS + MapLibre GL):** Renders real-time trajectory cones, active fires, registered institutional assets, and real-time execution logs directly from DynamoDB via Amazon API Gateway.

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4. Detailed Feature Specifications & User Stories

4.1 Feature 1: Fire Hotspot & Smoke Trajectory Vectoring

- **Description:** Processes active thermal hotspots and projects vector-based dispersion trajectories based on atmospheric wind patterns.
- **User Story:** *As an environmental safety director, I want to see calculated smoke dispersion paths from farm fires in real time so that I know exactly when hazardous air will reach my district.*
- **Technical Implementation:** Uses Python-based geometry processing in Lambda to generate polygon cones along wind vectors ($\vec{V} = u\hat{i} + v\hat{j}$), weighted by satellite thermal power (MW).

4.2 Feature 2: Institutional Vulnerability Mapping

- **Description:** Geofence registry mapping critical infrastructure and vulnerable population density points across Delhi NCR.
- **User Story:** *As a school principal, I want my campus registered in the system so that our facility automatically receives localized risk alerts before ambient air reaches dangerous levels.*

- **Technical Implementation:** DynamoDB spatial queries using geohash indexes to match projected dispersion polygons against registered institutional coordinates.

4.3 Feature 3: Autonomous Mitigation Advisory Engine

- **Description:** Bedrock-driven reasoning engine that translates trajectory metrics into targeted operational actions.
- **User Story:** *As a construction site manager, I want clear operational instructions rather than raw numbers so my team can adjust work shifts safely.*
- **Technical Implementation:** Enforced JSON schema generation via Amazon Bedrock with system-level safety guardrails.


```
{

  "institution_id": "SCH-DEL-0492",

  "facility_type": "School",

  "predicted_arrival_utc": "2026-10-09T08:30:00Z",

  "impact_severity": "CRITICAL",

  "recommended_actions": [

    "Cancel all outdoor activities and physical education classes scheduled after 08:00 AM.",

    "Switch central ventilation/HVAC systems to indoor air recirculation mode at 07:30 AM.",

    "Distribute N95 masks to student transport staff and outdoor security personnel."

  ],

  "confidence_score": 0.94

}
```

4.4 Feature 4: One-Click Simulated Smoke Surge & Emergency Broadcast Demo

- **Description:** A dedicated control trigger built into the dashboard interface to demonstrate end-to-end cloud execution live in under 3 seconds.
- **User Story:** *As a hackathon judge, I want to trigger a synthetic fire hotspot so I can observe the immediate end-to-end cloud ingestion, AI evaluation, and SMS alert delivery.*
- **Technical Implementation:** Invokes an Amazon API Gateway endpoint that feeds a simulated NASA FIRMS thermal anomaly directly into the Lambda evaluator pipeline.

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5. 72-Hour Sprint Execution Plan (Oct 8 – Oct 11, 2026)

OCT 8 (DAY 1) OCT 9 (DAY 2) OCT 10 (DAY 3) OCT 11 (DAY 4)

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| * AWS Infrastructure Setup | | * Ingestion Microservices | | * Interactive Map Frontend | | * 3-Min Video Recording |

| * DynamoDB Table Schemas | | * Bedrock Prompting Pipeline| | * Real-Time Dashboard UI | | * README & Architecture |

| * API Verification | | * Mock Alert Dispatchers | | * End-to-End Edge Testing | | * Final Submission Check-in|

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Phase 1: Infrastructure & Schema Setup (Day 1 — Oct 8)

- **09:00 - 12:00:** Initialize AWS SAM / CDK template project repository; setup IAM roles with least-privilege policies for Lambda, Bedrock, DynamoDB, and SNS.
- **12:00 - 15:00:** Validate NASA FIRMS API keys and Open-Meteo wind data endpoints using standalone Python scripts.
- **15:00 - 18:00:** Design DynamoDB table structures (`VayuSync_Hotspots`, `VayuSync_Institutions`, `VayuSync_AlertLogs`) and establish partition/sort keys.
- **18:00 - 21:00:** Deploy core AWS S3 bucket and SNS Topic infrastructure; confirm deployment pipeline using AWS CLI.

Phase 2: Core Microservices & Intelligence Engine (Day 2 — Oct 9)

- **09:00 - 13:00:** Build `VayuSync-IngestionEngine` Lambda function; implement wind vector calculation logic and trajectory cone polygon generation.
- **13:00 - 17:00:** Develop `VayuSync-EvaluatorEngine` integration with Amazon Bedrock; engineer system prompts to guarantee strict JSON output structures.
- **17:00 - 20:00:** Wire Amazon SNS and webhook notification services; configure email/SMS test endpoints and build simulated WhatsApp response mocks.
- **20:00 - 22:00:** Validate end-to-end backend event execution from EventBridge triggers to SNS dispatch via CloudWatch logs.

Phase 3: Frontend Dashboard & Integration (Day 3 — Oct 10)

- **09:00 - 13:00:** Scaffold React + Tailwind CSS dashboard; integrate MapLibre GL map view with custom layers for fire hotspots and trajectory polygons.
- **13:00 - 17:00:** Connect frontend directly to API Gateway endpoints to fetch live institutional alert logs, active trajectory cones, and system statuses.
- **17:00 - 19:00:** Build the "One-Click Simulated Smoke Surge" trigger interface on the dashboard for live judging execution.
- **19:00 - 22:00:** Conduct end-to-end edge testing, fix latency bottlenecks, verify overall system stability, and review project artifacts.

Phase 4: Media, Documentation & Final Submission (Day 4 — Oct 11)

- **09:00 - 12:00:** Script and record the strict 3-minute project demonstration video (showing architecture, live trigger, map visualization, and real-time SMS delivery).
- **12:00 - 14:00:** Finalize GitHub Repository documentation: craft README with architecture diagrams, deployment instructions, setup guides, and Bedrock prompt specs.
- **14:00 - 16:00:** Perform final AWS CloudWatch cleanup, double-check Free Tier compliance limits, and complete submission check-in.

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6. Judging Rubric Alignment & Winning Strategy

6.1 Matrix Alignment

Judging Criterion	VayuSync Strategic Advantage	Demonstration Proof Point
Real-World Impact	Moves beyond passive tracking to trigger automated, preventative operational changes 6–12 hours before smoke reaches high-density areas.	Show automated outdoor recess cancellation for a registered school district before local PM2.5 levels spike.
Depth of AWS Integration	Fully leverages 6 distinct AWS cloud services in a clean, event-driven serverless architecture.	Live AWS CloudWatch telemetry showing EventBridge $\rightarrow$ Lambda $\rightarrow$ Bedrock $\rightarrow$ SNS execution flow.
Demo Excellence	Zero slide deck dependence; uses a live, interactive React map equipped with a instant synthetic smoke surge trigger.	Trigger synthetic hotspot; show live trajectory generation, Bedrock

Judging Criterion	VayuSync Strategic Advantage	Demonstration Proof Point
		processing, and real-time SMS alert in < 3 seconds.
Code Quality & Architecture	Modular AWS SAM/CDK infrastructure with explicit IAM policies, clean code structure, and complete setup guides.	Public GitHub repository featuring clean microservice separation, setup instructions, and architecture diagrams.

6.2 The 3-Minute Winning Pitch Script Structure

1. **The Hook (0:00 - 0:35):** Show standard red AQI dashboards. "Traditional dashboards tell you when your lungs are already full of smoke. VayuSync tells institutions what to do *6 hours before* the smoke arrives."
2. **The Live Trigger Demo (0:35 - 1:35):** Click "Simulate Punjab Fire Hotspot". Show MapLibre dynamically drawing the 6-hour wind vector dispersion cone directly targeting North Delhi schools. Show the live phone screen receiving the automated SNS SMS alert.
3. **AWS Architecture Walkthrough (1:35 - 2:20):** Highlight the serverless chain: EventBridge triggers Lambda ingestion, trajectory polygon calculation, DynamoDB spatial matching, and Amazon Bedrock (Claude 3 Haiku) generating structured operational advisories.
4. **Targeted Impact & Scale (2:20 - 3:00):** Explain how VayuSync costs under $1/month to monitor all of Delhi NCR on AWS Free Tier while automating preemptive safety protocols across schools, hospitals, and outdoor work sites.